Decisions and Repetition · AS

U05e · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Fresh authored questions. Pseudocode only. Assume valid numeric input where stated. Use the contracts printed in the question.

1. Ours: independent decisions

```
DECLARE Score : INTEGER
Score ← 84
IF Score >= 60 THEN
    OUTPUT "Pass"
ENDIF
IF Score >= 80 THEN
    OUTPUT "Star"
ENDIF
```

State the output at Score 84 [2]. State the output if Score starts at 70 [1]. Explain why both IF bodies can run [1]. [4]

2. Ours: the menu

```
DECLARE Choice : INTEGER
INPUT Choice
CASE OF Choice
    1 : OUTPUT "Map"
    2 : OUTPUT "Torch"
    OTHERWISE : OUTPUT "Help"
ENDCASE
```

State the output for inputs 2 and 8 [2]. Explain what OTHERWISE does [1]. State the keyword that closes this selection [1]. [4]

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3. Ours: count the visited values

For each FOR header below, list every counter value visited, or write none [1 each].
(i) FOR I ← 2 TO 11 STEP 3
(ii) FOR I ← 8 TO 2 STEP -2
(iii) FOR I ← 4 TO 4
(iv) FOR I ← 5 TO 2 STEP 1 [4]

4. Ours: the first trace

```
DECLARE N, Total : INTEGER
N ← 0
Total ← 1
REPEAT
    N ← N + 1
    Total ← Total + 2 * N
UNTIL N = 3
OUTPUT Total
```

Draw a trace table with N, Total and the UNTIL result after each body execution. Give all three complete rows [3]. State the final output [1]. [4]

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5. Ours: choose and justify

(i) A display must print all 18 labels. Choose a loop and justify it [2].
(ii) A payment balance may already be at least 20 before any new coin is entered. Choose a loop that avoids unnecessary input and justify it [2]. [4]

6. Ours: repair a repeat loop

```
DECLARE N : INTEGER
N ← 2
REPEAT
    OUTPUT N
    N ← N - 1
UNTIL N = 6
```

State the first two outputs [2]. Replace one line so that the program outputs 2, 3, 4, 5 and stops [1]. Explain why the repaired program does not output 6 [1]. [4]

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7. Ours: pen first: count the lanterns

6 marks, all completed on paper. A route has lanterns numbered 1 to 6. Count only lanterns whose number is at least 4. Output the accepted count once after the loop. No computer or separate code file is required. Plan before writing the instructions.

(a) Write numbered steps or draw a flowchart showing the counter, selection and final output. [2]

(b) Write declared pseudocode using FOR with an IF inside it. Initialise the accepted count correctly. [3]

(c) State the final output for the six lanterns. [1]

Above and beyond · outside the mark total. Starting from 1, repeatedly add the next odd number: 3, 5, 7, … . Trace five totals. What shape can you build from these totals, and why?

Additional space · label the question number.